网络关系与财务欺诈识别

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财务欺诈识别文献梳理

- 过往财务欺诈识别主要依赖企业自身信息
  - 会计变量（Bao et al., 2020; Bertomeu et al., 2021）
  - 文本信息（Brown et al., 2020; Liu et al., 2023）
  - 数据与文本（Craja et al., 2020; Campbell and Shang, 2022）
  - 多模态数据（Lu et al., 2025; Chang et al., 2025）
- 但企业并不是孤立存在的，企业和社会环境之间存在复杂的交互关系。
- 企业开始被放进网络中：
  - 产品网络（Product Network）
  - 供应链网络（Supply Chain Network）
  - 同业网络（Peer Network）
  - 股权网络（Ownership Network）
  - 知识网络（Knowledge Network）
  - 信息网络（Information Network）
  - 关联交易网络（Related-Party Transaction Network）

网络关系与财务欺诈识别

- 供应链网络中的财务欺诈识别 (Li et al., 2023, TAR)
- 关联交易网络和股权网络中的财务欺诈识别 (Du et al., 2025, MISQ)
- 产品网络中的财务欺诈识别 (Boone et al., 2025, JFQA)
- 同业网络中的财务欺诈识别 (Fan et al., 2026, MS)
- 信息网络中的财务欺诈识别 (Fan et al., 2025, Working Paper)
- 网络关系建模方法
  - 数据指标关联关系 (Li et al., 2023)
  - 文本相似度 (Boone et al., 2025; Fan et al., 2026)
  - 静态图优化 (Fan et al., 2025)
  - 图注意力网络动态特征提取 (Du et al., 2025)

Measuring Misinformation in Financial Markets

Jianqing Fan, Qingfu Liu, Yang Song, Zilu Wang (2025, *Working Paper*)

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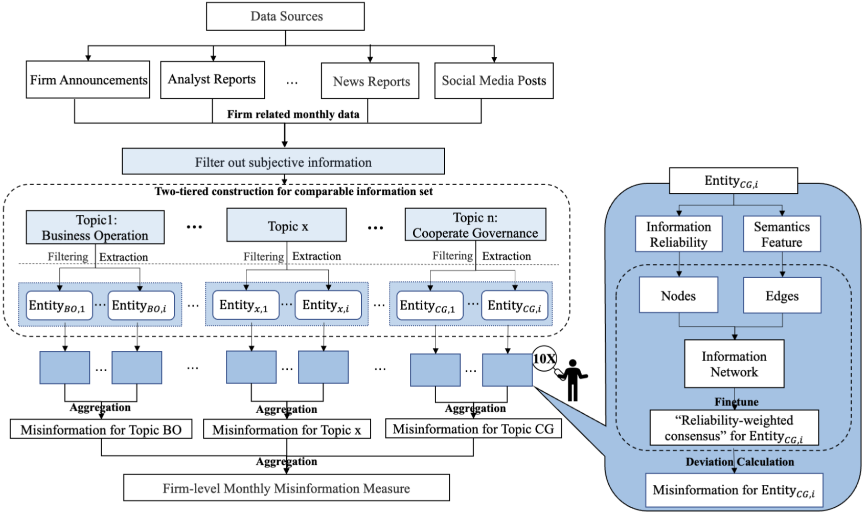
Overview


Data Sources
Firm Announcements
Analyst Reports
...
News Reports
Social Media Posts
Firm related monthly data
Filter out subjective information
Two-tiered construction for comparable information set
Topic1: Business Operation
Topic x
Topic n: Cooperate Governance
Filtering
Extraction
Entity$_{BO,1}$
Entity$_{BO,i}$
Entity$_{x,1}$
Entity$_{x,i}$
Entity$_{CG,1}$
Entity$_{CG,i}$
10X
Aggregation
Misinformation for Topic BO
Misinformation for Topic x
Misinformation for Topic CG
Aggregation
Firm-level Monthly Misinformation Measure
Entity$_{CG,i}$
Information Reliability
Semantics Feature
Nodes
Edges
Information Network
Finetune
"Reliability-weighted consensus" for Entity$_{CG,i}$
Deviation Calculation
Misinformation for Entity$_{CG,i}$

Motivation

- Challenge in measuring misinformation
  - Misinformation: misleading or factually incorrect information, refer to unintentional error and intentionally fabricated content (disinformation created and spread to deceive)
  - Measuring misinformation require evaluating statements against objective facts, but facts unobservable or infeasible to verify at scale.
  - Research limit due to asymmetric nature and lack of disclosure of genuine information.
- Generative AI for quantifying firm-level misinformation
  - Understanding capability for information labeling.
  - Efficiency and speed for information embedding.
  - Cost-effectiveness of few-shot framework for text classification and extraction.

Research Question

- How to measure misinformation in financial market?
- What drives misinformation?
- How misinformation impacts investor behavior, stock return, and firm-level risk?

Contribution

- Contribute to literature that identifying misinformation.
  - Prior: use expost rumor event or SEC enforcement action to identify fraudulent disclosure (Schmidt, 2020; Clarke et al., 2020); apply LIWC2015 dictionary to infer deception by word frequency (Kogan et al., 2023; Liu and Moss, 2025).
  - Limit: narrow subset of confirmed case; psychological cue rather than factual accuracy.
  - This paper: identify and quantify misinformation in rigorous and scalable way.
- Contribute to literature on financial textual analysis.
  - Prior: analyst reports, social media posts, news articles influence asset price (Jiang et al., 2019; Cohen et al., 2020; Beckmann et al., 2024).
  - Limit: these source exist biases skew perception and decision-making (Schmidt, 2020).
  - This paper: employ LLM and network approach to assess quality of textual information.

Framework for Measuring Misinformation

- REE framework
  - Factual state of a firm on a given dimension as an unobserved latent variable.
  - Each objectively verifiable statement as a noisy signal about latent state.
- Two guiding principles
  - **Information consistency**: signals internally coherent and corroborated across diverse sources more likely close to latent factual state.
  - **Wisdom of select crowds**: not all sources equally reliable, higher credible sources weighted more heavily.

Multi-source Firm-level Textual Corpus (2015-2022)

- Firm disclosure
  - 1,071,464 records of firm announcements, annual and quarterly reports, investor Q&A sessions from CSMAR and WIND
- News report
  - 3,327,580 news reports from Huike and CSMAR
  - 13 major media sources classified as official, semi-official, unofficial (An et al., 2022)
- Analyst report
  - 568,468 analyst reports from CSMAR, WIND, and Luobo Investment Research
- Social media
  - 285,080,647 posts from GUBA

Strip Away Subjectivity: Identify Objectively Verifiable Text

- Misinformation vs. Disagreement
  - Misinformation concern factual accuracy of objectively verifiable information.
  - Disagreement reflect differences in subjective interpretation and beliefs.
  - Forecast ultimately proved incorrect not misinformation, reflect differing view on future.
  - Subjective claim (inference opinion, emotional expression, forward-looking prediction) cannot find clear factual benchmark.
- Chunking segment
  - Segment all text by inherent paragraph structure and headings.
  - Further split if exceed 200 character limit and eliminate if shorter than 10 characters.
- LLM to classify text as objectively verifiable or subjective
  - Annotate random sample of 40,000 text by ChatGPT-4o and manually review.
  - Fine-tune BERT by adding binary classification layer on top.
  - Assign all text to class with higher probability based on BERT embedding.

From Clutter to Clusters: Categorize Text into Comparable Subset

- Topic-entity tuple (o, e)
  - Label text with coarse-grained topic label (o) and fine-grained entity label (e)
- Topic classification
  - Predefine 8 topics: business operations (BO), financial performance (FP), risk management (RM), corporate governance (CG), environmental responsibility (ER), social responsibility (SR), human capital (HC), R&D innovation (RD)
  - Encode texts by ERNIE (BERT variant developed by Baidu)
  - Match to topic by USM (share space for label and text through Direct-Token-Link)
- Entity extraction
  - Extract entity in a unsupervised way by fine-tuning USM on ERNIE embedding.
  - Group synonymous entity for canonicalization by hierarchical clustering.

Extracted Topics and Entities

(a) Financial Performance

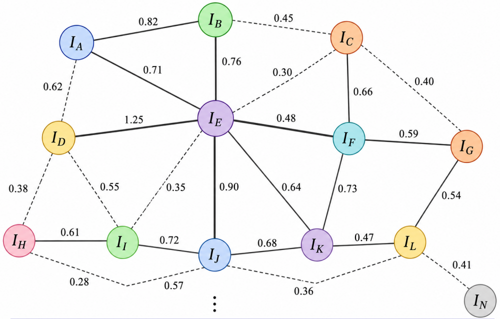
(b) Risk Management

(c) Environmental Responsibility

(d) Human Capital

Find Benchmark: Build Information Network

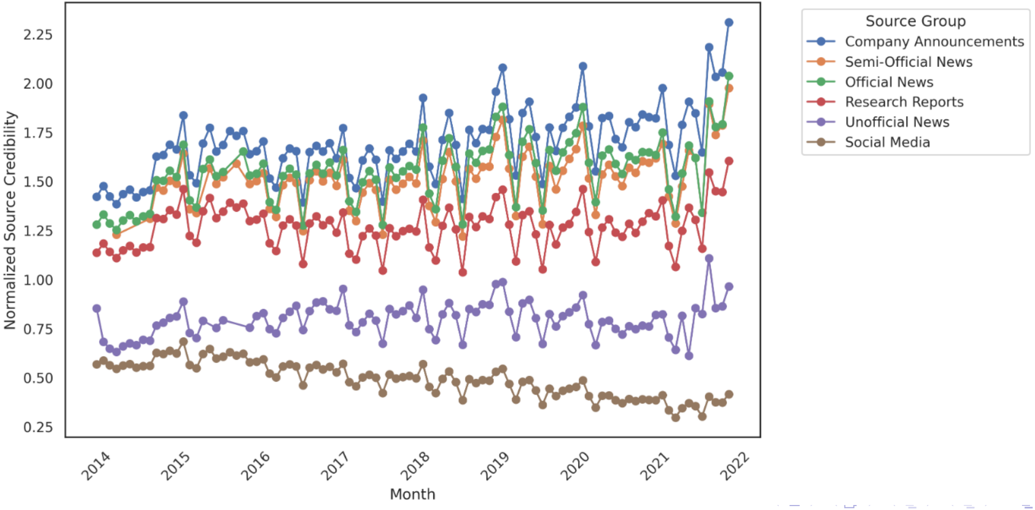
- Constructing information network
  - Undirected information network $\mathcal{G} = \langle \mathcal{V}, \mathcal{E} \rangle \Rightarrow$ information set I associated with (o, e)
  - Node $\mathcal{V} \Rightarrow$ individual piece of information in I
  - Edge $\mathcal{E} \Rightarrow$ cosine similarity between information i and j $s(\mathbf{x}_i, \mathbf{x}_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\|_2 \times \left\|\mathbf{x}_j\right\|_2}$
  - Edge weight matrix $\mathbf{B} = (b_{ij})_{|I|\times|I|} \Rightarrow b_{ij} = 1 + s(\mathbf{x}_i, \mathbf{x}_j)$



Find Benchmark: Measuring Reliability of Information

- Reliability = Source credibility + Content coherence + Cross-source corroborate
  - Recognize varying and dynamic nature of source credibility.
  - Identify potentially unreliable or anomalous information through content coherence.
  - Mitigate source-specific bias through network smoothing.
- Source credibility: $c_{s,t} = c_{s,t-1} \times e^{-\alpha MISI_{s,t-1}}$
  - Initialize by ChatGPT-4 and adjust based on propensity to spread misinformation.
- Content coherence: $l_i = \frac{1}{n} \sum_{j=1}^{n} \log P(\text{word}_j \mid \text{word}_{<j})$
  - Infer through formation probability assigned by GPT given previous context.
- Cross-source corroborate: $\min_{\tilde{\mathbf{w}}} \mathcal{J}(\tilde{\mathbf{w}}) = \min_{\tilde{\mathbf{w}}} \eta \left\|\tilde{\mathbf{w}} - \mathbf{w}\right\|^2 + \sum_{i,j=1} b_{ij} \left| \frac{1}{\sqrt{f_i}} \tilde{w}_i - \frac{1}{\sqrt{f_j}} \tilde{w}_j \right|^2$
  - Balance two graph optimization goals: (1)preserve initial reliability assessment; (2)assign similar reliability weights to semantically similar information.

Credibility Weights over Time by Information Sources



Measure Misinformation: Aggregate Deviation from Benchmark

- Extracting benchmark
  - Estimate latent factual state as semantic center of information
  - Optimization problem: $\mathbf{x}^* = \arg\min_{\mathbf{x}} \sum_{i \in I} \tilde{w}_i \cdot d(\mathbf{x}_i, \mathbf{x}) = \arg\min_{\mathbf{x}} \sum_{i \in I} \tilde{w}_i \cdot (1 - s(\mathbf{x}_i, \mathbf{x}))$
  - Balance information reliability and information volume
- Measuring misinformation
  - Hierarchical aggregation: $Piece \xrightarrow{\text{mean}} Entity \xrightarrow{\text{equal weight}} Topic \xrightarrow{\text{coverage weight}} Firm$
  - Piece-level: $MISI_i^{piece} = d(\mathbf{x}_i, \mathbf{x}^*) = 1 - \frac{\mathbf{x}_i \cdot \mathbf{x}^*}{\|\mathbf{x}_i\|_2 \times \|\mathbf{x}^*\|_2}$
  - Entity-level: $MISI_e^{entity} = \frac{1}{|\mathcal{I}^e|} \sum_{i \in \mathcal{I}^e} MISI_i^{piece}$
  - Topic-level: $MISI^{topic} = \sum_{e \in E^o} MISI_e^{entity}$
  - Firm-level: $MISI^{firm} = \frac{1}{\sum_{o \in \mathcal{O}} |\mathcal{I}^o|} \sum_{o \in \mathcal{O}} |\mathcal{I}^o| \cdot MISI_o^{topic}$

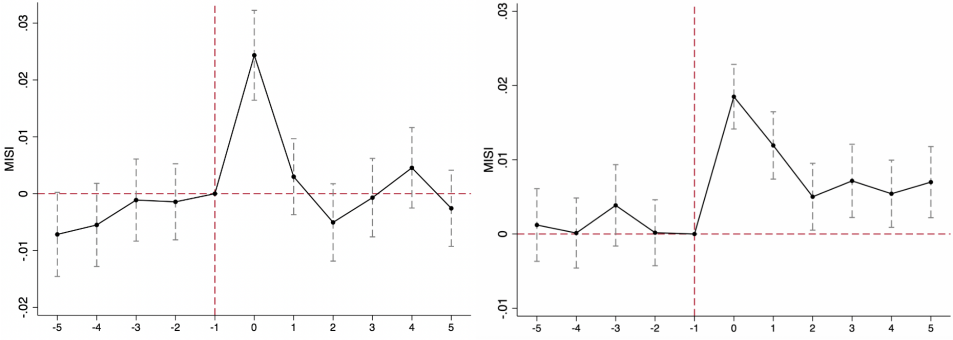
Misinformation during Misinformation Event

- Publicly disclosed misinformation event
  - Company announcements of rumor clarifications obtained from CNRDS
  - Penalty announcements on firm fraud issued by CSRC
  - Keyword filter: 伪造记录, 重大差异, 误导性陈述, 与事实不符
  - 1,079 misinformation events involving 537 firms
- Text validation
  - Candidate-misinformation set: pieces most similar to false claims in official document.
  - Baseline set: pieces released by same firm in same month of previous year.

	Candidate-misinformation set	Baseline set
Panel A: strategy 1		
Number of pieces	181	112,360
Mean MISI	0.1552	0.0869
Difference in means	0.0683***	

Dynamic Pattern in MISI around Misinformation Event

- $MISI_{i,t} = \beta_0 + \sum_{j=2}^{5} \gamma_j I_{i,t-j} + \sum_{k=0}^{5} \alpha_k I_{i,t+k} + \eta \times Firm\ Characteristics + FEs + \epsilon_{i,t}$



(a) rumor event **(b)** fraud event

- Rumor tend to be addressed and clarified quickly
- Fraud lead to prolonged period of misinformation until eventual disclosure

Relationship between Misinformation and Disagreement

- DISA: residual corpus filter out all objectively verifiable statements
  - (1)Group into comparable set; (2)Calculate pairwise similarity using embedding; (3)Compute standard deviation of similarity within set; (4)Aggregate at firm-level

Panel A: misinformation → disagreement

	Dep. Variable: $DISA_{t+1}$		Dep. Variable: $DISA_{t+2}$		Dep. Variable: $DISA_{t+3}$	
	(1)	(2)	(3)	(4)	(5)	(6)
$MISI_t$	0.0046***	0.0038***	0.0039***	0.0034**	0.0024*	0.0018
	(0.0014)	(0.0014)	(0.0013)	(0.0014)	(0.0013)	(0.0014)
$DISA_t$	0.1474***	0.1411***	0.0883***	0.0837***	0.0686***	0.0653***
	(0.0031)	(0.0032)	(0.0026)	(0.0027)	(0.0025)	(0.0025)
$DISA_{t+1}$			0.1286***	0.1251***	0.0764***	0.0735***
			(0.0027)	(0.0027)	(0.0025)	(0.0026)
$DISA_{t+2}$					0.1192***	0.1165***
					(0.0026)	(0.0027)

Panel B: disagreement → misinformation

	Dep. Variable: $MISI_{t+1}$		Dep. Variable: $MISI_{t+2}$		Dep. Variable: $MISI_{t+3}$	
	(1)	(2)	(3)	(4)	(5)	(6)
$DISA_t$	0.0037	0.0033	0.0045	0.0035	0.0032	0.0026
	(0.0037)	(0.0037)	(0.0036)	(0.0036)	(0.0035)	(0.0036)
$MISI_t$	0.1001***	0.0974***	0.0608***	0.0576***	0.0511***	0.0492***
	(0.0029)	(0.0029)	(0.0025)	(0.0025)	(0.0026)	(0.0027)
$MISI_{t+1}$			0.0938***	0.0911***	0.0571***	0.0541***
			(0.0027)	(0.0027)	(0.0024)	(0.0025)
$MISI_{t+2}$					[illegible]	0.0880***

Firm Characteristics and Misinformation

- Balance sheet, financial situation, governance structure

	Dep. Variable: MISI			
	(1)	(2)	(3)	(4)
Debt	0.0003*			0.0004**
	(0.0002)			(0.0002)
Size	0.0041***			-0.0098***
	(0.0004)			(0.0036)
Cash	-0.0033			-0.0029
	(0.0024)			(0.0025)
RevGrowth	-0.0012***			-0.0011***
	(0.0002)			(0.0002)
FinRisk		0.0261***		0.0357***
		(0.0041)		(0.0052)
FinConstraint		0.0038***		0.0121***
		(0.0003)		(0.0029)
IndBoard			-0.0043***	-0.0038**
			(0.0014)	(0.0016)
Institution%			-0.0007***	-0.0007**
			(0.0002)	(0.0003)
ShareCon			0.0001	0.0001
			(0.0001)	(0.0001)

Major Corporate Events and Misinformation

- Financing operation, corporate split, merger and acquisition

	Panel A: financing-related actions			
	Dep. Variable: MISI			
	(1)	(2)	(3)	(4)
Actions	0.0014***	0.0013***	0.0012***	0.0013***
	(0.0001)	(0.0001)	(0.0001)	(0.0001)
	Panel B: including other corporate actions			
	Dep. Variable: MISI			
	(5)	(6)	(7)	(8)
$\text{Actions}^{\text{Com}}$	0.0009***	0.0009***	0.0009***	0.0009***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)

Misinformation and Trading Activity

	Panel A: MISI and trading volume					
	Dep. Variable: $AdjTrade_t$			Dep. Variable: $AdjTrade_{t+1}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$MISI_t$	0.6419***	0.4213***	0.2883***	0.4751***	0.3071***	0.0243*
	(0.0208)	(0.0209)	(0.0139)	(0.0199)	(0.0204)	(0.0129)
$AdjTrade_t$						0.6401***
						(0.0043)

	Panel B: MISI and trade type					
	OIB_t^{Small}	OIB_t^{Medium}	OIB_t^{Large}	OIB_{t+1}^{Small}	OIB_{t+1}^{Medium}	OIB_{t+1}^{Large}
	(1)	(2)	(3)	(4)	(5)	(6)
$MISI_t$	0.0074***	0.0184***	-0.0475**	0.0088***	0.0070**	-0.0075
	(0.0028)	(0.0034)	(0.0195)	(0.0028)	(0.0033)	(0.0199)
OIB_t^{Small}				0.1862***		
				(0.0041)		
OIB_t^{Medium}					0.1959***	
					(0.0042)	
OIB_t^{Large}						0.0395***
						(0.0035)

- Misinformation prompt more frequent and aggressive trading activity.
- Small and medium trades net buying, large trades net selling

Misinformation and Price Dynamics

Panel A: MISI and mispricing

	Dep. Variable: MISVAL			Dep. Variable: $\mathbb{I}\{\text{MISVAL} > 0\}$		
	(1)	(2)	(3)	(4)	(5)	(6)
MISI	0.4138***	0.0648***	0.0724***	0.1561***	0.0351	0.0637**
	(0.0448)	(0.0232)	(0.0228)	(0.0423)	(0.0250)	(0.0257)

Panel B: MISI and stock return

	Dep. Variable: Return_t		Dep. Variable: $\text{Return}_{t+1,t+3}$		Dep. Variable: $\text{Return}_{t+1,t+6}$	
	(1)	(2)	(3)	(4)	(5)	(6)
MISI_t	0.0239***	0.0261***	-0.0687***	-0.0445***	-0.1328***	-0.0735***
	(0.0063)	(0.0064)	(0.0112)	(0.0116)	(0.0174)	(0.0182)
Return_t				-0.1475***		-0.1973***
				(0.0046)		(0.0060)

- Misinformation contributes to mispricing.
  - Firm with higher misinformation more likely overvalued relative to fundamental.
- Pricing distortion followed by return reversal.
  - Speculator incentive to spread positive misinformation to inflate prices, then reversal.

Misinformation, Maximum Drawdown, and Stock Crash Risk

	Panel A : MISI and maximum drawdown			
	Drawdown_{t+1}	$\text{Drawdown}_{t+1,t+3}$	$\text{Drawdown}_{t+1,t+6}$	$\text{Drawdown}_{t+1,t+12}$
	(1)	(2)	(3)	(4)
MISI_t	0.0241***	0.0345***	0.0485***	0.0527***
	(0.0032)	(0.0043)	(0.0062)	(0.0084)
Drawdown_t	-0.0782***	-0.1081***	-0.1325***	-0.1481***
	(0.0028)	(0.0035)	(0.0047)	(0.0056)
	Panel B: MISI and stock crash risk			
	NCSKEW_{t+1}	$\text{NCSKEW}_{t+1,t+3}$	$\text{NCSKEW}_{t+1,t+6}$	$\text{NCSKEW}_{t+1,t+12}$
	(1)	(2)	(3)	(4)
MISI_t	0.0852*	0.1850***	0.1961***	0.0877*
	(0.0446)	(0.0560)	(0.0478)	(0.0494)
NCSKEW_t	-0.0643***	-0.0744***	0.2065***	-0.0445***
	(0.0037)	(0.0018)	(0.0016)	(0.0012)

- Misinformation predict stock downside risk over subsequent year.

Discussion

- From static benchmark to dynamic latent state
  - This paper implicitly segments company's information flow month by month
  - Company's actual state is fixed each month and that a static benchmark exists
  - Company's status is dynamically evolving.
  - Information from different times may all be true, but they correspond to different states.
  - State-space model: $\mathbf{z}_t = f(\mathbf{z}_{t-1}) + \mathbf{u}_t$; $\mathbf{x}_t = g(\mathbf{z}_t) + \mathbf{v}_t$
- From consensus truth to causal truth
  - This paper assume at least regulated source anchor truth and error not fully correlated.
  - Assumption break when credible sources simultaneously disseminate misinformation.
  - External fact anchors: Financial data, supply chain data, patent data, recruitment data
  - $Truth = f(Text, Accounting, Operations, Market)$

Discussion

- Heterogeneous graph neural networks
  - Different node types: companies, media, analysts, KOLs, retail investors
  - Different edge types: quoting, forwarding, tracking, commenting
  - Graph Transformer
- Strategic misinformation
  - Some deviation due to random error.
  - Some deviation strategic misdirection (selection, positioning, emphasis, dispersion).
  - LLM Reasoning identify inform, persuade, manipulate
- Financial hallucination
  - View firm disclosures as generative process
  - Firm $\rightarrow$ language model, disclosure $\rightarrow$ generated text, real State θ_t $\rightarrow$ ground truth
  - $\text{Hallucination}_t = d(\text{Disclosure}_t, \theta_t)$

中国市场误信息识别

- 语料数据（2023.01 - 2024.12）
  - 年报段落文本爬取自巨潮资讯
  - 公司公告文本爬取自巨潮资讯
  - 新闻文本爬取自新浪财经
  - 分析师报告爬取自东方财富网站
  - 股吧文本爬取自东方财富网站
- 文本预处理
  - 文本分段：根据段落结构分段，超过 200 字的进一步拆分，不足 10 字的剔除。
  - 客观可验证文本识别：使用 Qwen-3 对文本进行标注
  - 主题-实体元组标注：预定义 8 个主题，使用 Qwen-3 进行主题分类和实体抽取。

主题-实体元组标注 Prompt

```python
PROMPT = """
你是一名金融专家，擅长分析和理解金融市场中的各类信息。文本来源包括公司公告、研究报告、新闻报道和社交媒体等。你的任务是分析给定的中文文本内容，并完成以下步骤：

1）判断该文本是否构成完整句子或完整段落，即语义是否连贯。
如果语义连贯，请继续完成后续任务；
如果语义不连贯，请直接按照以下格式回答，并停止后续任务：
主题：其他
实体：语义不连贯

2）将文本分类到一个或多个与具体公司行为相关的主题中。
可选主题只能包括以下类别：财务绩效；经营业务；公司治理；人力资本；社会责任；环境责任；风险管理；研发创新；其他。

请注意：
a. 选择的主题必须反映公司的具体行为或状态，不能过于宽泛。
如果文本内容过于笼统，请分类为 其他。
b. 主题名称必须严格使用上述给定类别，不得自行创造新类别。
c. 应根据文本的核心内容进行分类。

3）从文本中提取核心被描述实体，例如产品、业务、项目、人物、部门、事件等。
立场只能为 积极、消极或中性。
主题与可选实体范围如下：
财务绩效 可选实体包括：每股收益、收入、营业收入、利润、净利润、毛利润、归属于股东的利润、现金流、成本、财务费用、销售收入、市值、估值、资产、净资产、利润率、经营绩效、企业绩效、年报、财务报告。
经营业务 可选实体包括：产品、服务、行业、市场、核心业务、项目、生产能力、产能、业务发展、市场份额、客户、销售、订单、渠道、生产效率、经营收入、成本、利润、并购重组、产品结构、产品质量、业务规模、市场需求。
公司治理 可选实体包括：股东、股权、股份、股票、股票质押、机构投资者、管理层、管理人员、董事会、监事会、公司秘书、公司治理、内部控制、信息披露、公告、重组并购、注册资本、股本、激励计划、员工持股、股东大会。
人力资本 可选实体包括：员工、人才、核心员工、员工持股计划、激励机制、薪酬、工资、管理能力、专业人员、技术人员、行政费用、销售费用、人力资源、员工结构、员工素质、人才团队、用工成本、员工稳定性。
社会责任 可选实体包括：社会责任、责任、职业、社会效益、民生、环境、项目、建设、政策、国家、世界、经济、就业、安全生产、公益、社区、员工权益、客户权益、产品安全、服务质量。
环境责任 可选实体包括：环境、环境保护、环保政策、环境标准、环境问题、环保设备、环保投入、生产过程、能源、原材料、排放、污染、生态、绿色发展、节能减排、环境治理、环保监管、生产能力、成本、产品价格。
风险管理 可选实体包括：风险、管理、机构、资金、股票质押、信息披露、项目可行性、成本、市场竞争、重大股东、承诺、项目、公告、控制、重组、并购、资产、价格、法律风险、经营风险、财务风险、市场风险。
研发创新 可选实体包括：技术、研发、研究与开发、研发投入、研发费用、科技、技术研发、技术创新、知识产权、专利、芯片、产品、产品结构、生产能力、设备、项目、创新能力、科研人员、技术人员、产品市场、产品质量。

请注意：
a. 如果文本中包含多个被描述实体，请用英文分号 ; 分隔。
b. 如果某一项未提及，请填写 未提及。

请严格按照以下格式回答：
Topic: [你的答案]
Entity: [你的答案]
Content: [你的答案]
Stance: [积极/消极/中性]
"""
```

Python

构建信息网络

- 使用 Qwen-3 生成文本片段的语义向量表示 $\mathbf{x}_i$
- 基于余弦相似度计算文本片段之间的语义关联，构建边权矩阵 $\mathbf{B} = (b_{ij})$
- 初始化信息可靠性权重：

$$w_i^{(0)} = \begin{cases} 0.9, & \text{年报、公司公告} \\ 0.8, & \text{分析师报告} \\ 0.6, & \text{新闻文本} \\ 0.3, & \text{股吧文本} \end{cases}$$

- 计算 Cross Entropy Loss，得到文本内容连贯性得分 $l_i = -\mathcal{L}_i$
- 综合来源可信度与内容连贯性，更新信息可靠性权重 w_i
- 通过图优化，对相似文本的信息可靠性权重进行平滑

文本验证

- 利用 CSMAR 数据库中 2024 年的违规事件进行文本验证，识别到 132 个和误信息相关且发生日期明确的事件
- 提取出其中提及误信息的相关陈述，语义检索来近似目标片段，取前五条最相关的文本作为候选误信息文本
- 构建基准文本集为同一公司前一年同月发布的文本

表: 基于 CSMAR 2024 年违规事件的文本验证

	候选误信息集合	基准信息集合
文本数量	327	14,508
平均 MISI	0.134	0.071
均值差异	0.063**	

MISI 与误信息违规事件

	被解释变量：误信息违规事件		
	(1)	(2)	(3)
MISI	0.0091*	0.0112**	
	(0.0053)	(0.0049)	
ΔMISI			0.0057**
			(0.0028)
控制变量	NO	YES	YES
公司固定效应	YES	YES	YES
月份固定效应	YES	YES	YES
行业 × 年份固定效应	YES	YES	YES
样本量	54,146	52,379	47,845
R^2	0.039	0.043	0.041

MISI 围绕误信息违规事件动态模式